

Speech-Silence Segmentation Using Short-Time DSP Features

Digital Signal Processing - Midterm Project

Three teacher-aligned baselines and three DSP extensions

Group members: [Student name - ID] | [Student name - ID] | [Student name - ID]

Dataset: four training recordings and four independent test recordings

1. Problem and required output

The input is a recorded speech signal. The output is the estimated start and end of the speech region. One Run All must process the four test recordings and show one figure per recording: waveform, normalized short-time energy, reference boundaries, predicted boundaries, F0 interpretation track, and boundary error in milliseconds.

All methods use 20 ms frames, a 10 ms hop, and the required rule that a true silence interval lasts at least 200 ms. Shorter internal zero runs are bridged as false silence gaps.

2. Data and experimental protocol

Split	Recordings	Role
Training	phone_F1, phone_M1, studio_F1, studio_M1	Estimate parameters
Validation	studio_M1 held out	Model selection without test data
Test	phone_F2, phone_M2, studio_F2, studio_M2	Final comparison only

The split is performed by complete recording. Neighboring frames from one utterance never cross the fit/validation boundary. After validation, the selected method is fitted again with all four training recordings. Test labels are used only for final metrics and red reference lines.

3. Common DSP pipeline

Normalized STE is $E[m] = \text{mean}(x_m[n]^2) / \max_k \text{mean}(x_k[n]^2)$. It is a strong baseline because voiced speech is normally more energetic than silence. Its main failure mode is weak unvoiced speech, whose energy may approach background noise.

The extensions add the FIR pre-emphasis filter $h[n] = [1, -0.97]$, zero-crossing rate, high-frequency energy ratio above 3 kHz, and log-energy. The filter $y[n] = x[n] - 0.97x[n-1]$ reduces low-frequency dominance and highlights rapid high-frequency changes. F0 is estimated by autocorrelation for visualization only; it is not used to determine boundaries.

4. TT1 - global threshold search

4.1 Teacher-aligned baseline

Binary search learns one global normalized-STE threshold by minimizing training boundary error. Inference applies the threshold and then the 200 ms silence rule.

4.2 DSP extension

A small logistic classifier combines log-STE, pre-emphasized energy, ZCR, and high-frequency energy ratio. Training loss is stored in the notebook, and no external machine-learning package is used.

TT1 - Baseline: four test recordings

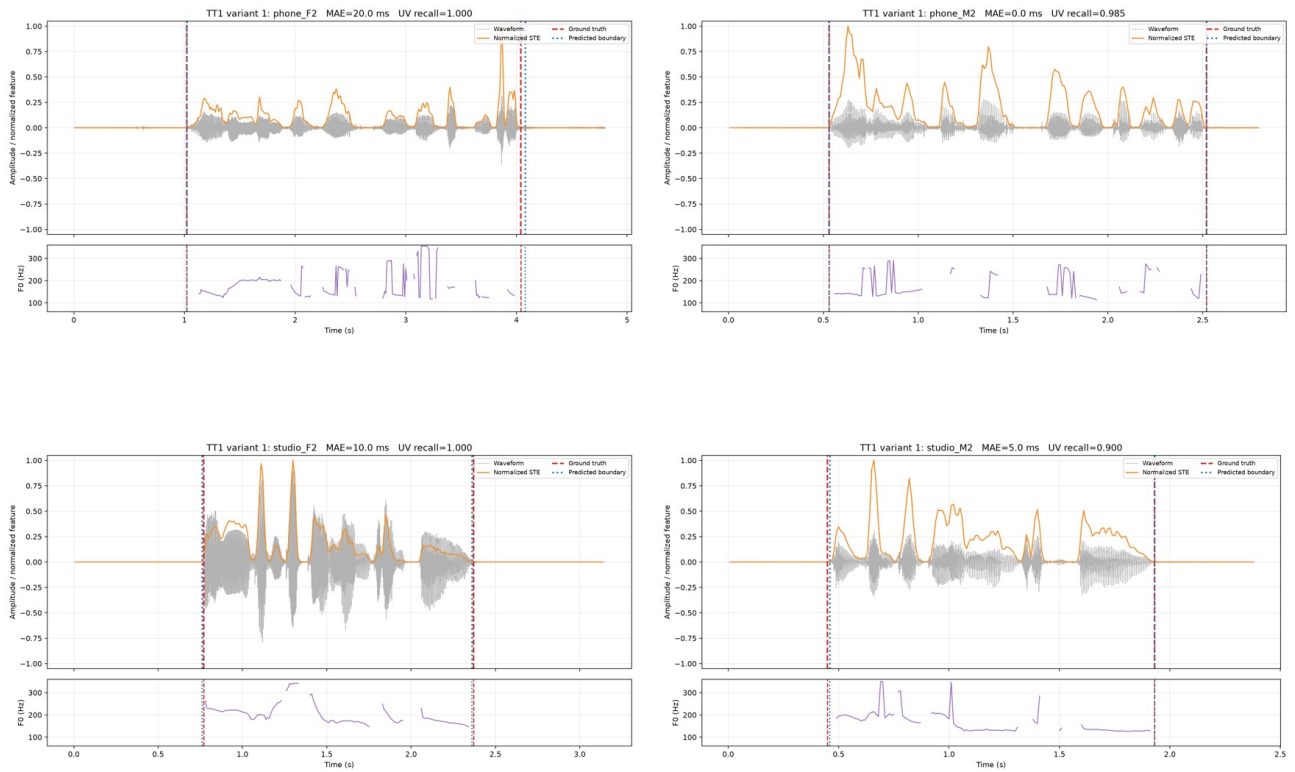


Figure 1. TT1 teacher baseline on all four test recordings.

TT1 - DSP extension: four test recordings

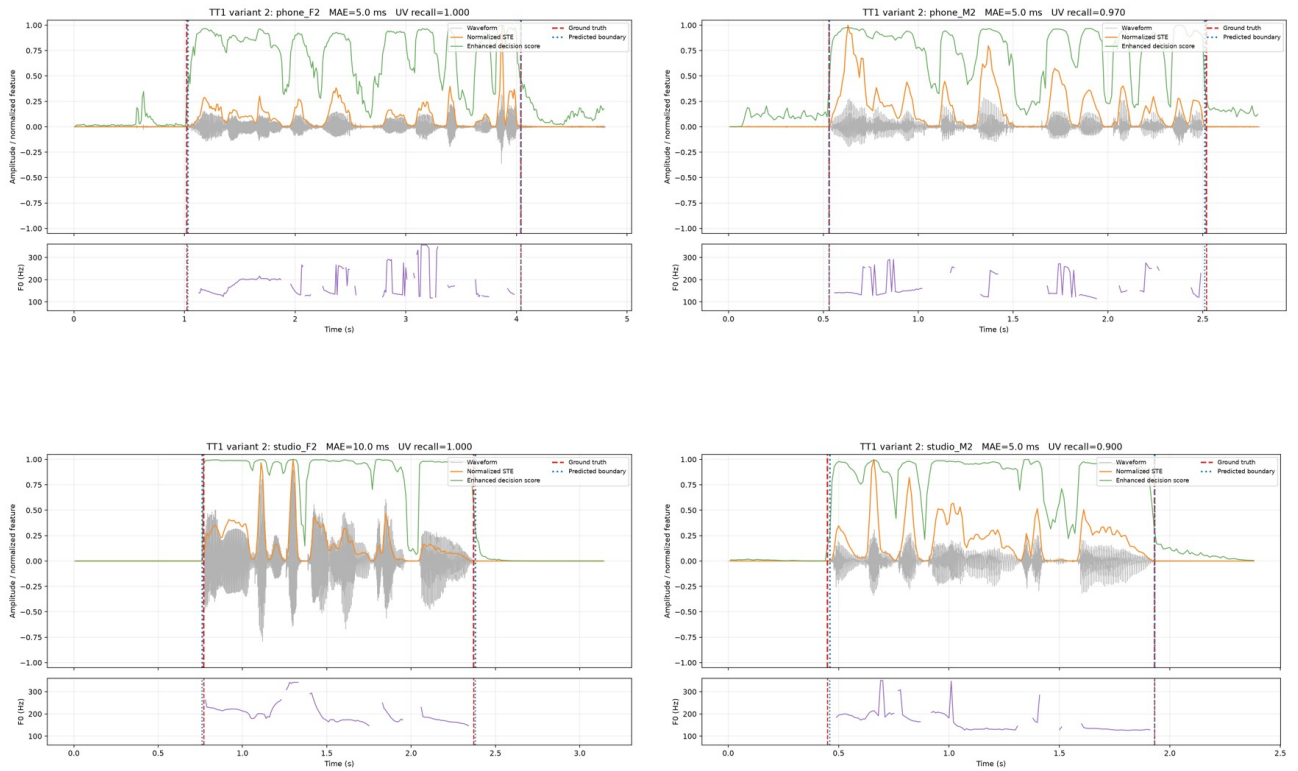


Figure 2. TT1 DSP extension on all four test recordings.

TT1 gives the best boundary result. The extension decreases mean MAE from 8.75 ms to 6.25 ms and mean RMSE from 11.34 ms to 7.80 ms. Frame F1 increases slightly. UV recall changes from 0.971 to 0.968, which is practically similar on this small set.

5. TT2 - histogram method

5.1 Teacher-aligned baseline

The one-dimensional histogram $H[n]$ is computed from normalized STE. The low-energy mode represents silence and the higher-energy mode represents speech; a weighted point between them becomes the adaptive threshold.

5.2 DSP extension

The extension uses the joint histogram $H[\text{normalized STE}, \text{ZCR}]$. Every bin stores smoothed speech and silence counts learned from the training labels. The decision is their log-count ratio. A training-derived energy floor stops very quiet background frames from being accepted only because of a noisy ZCR bin.

TT2 - Baseline: four test recordings

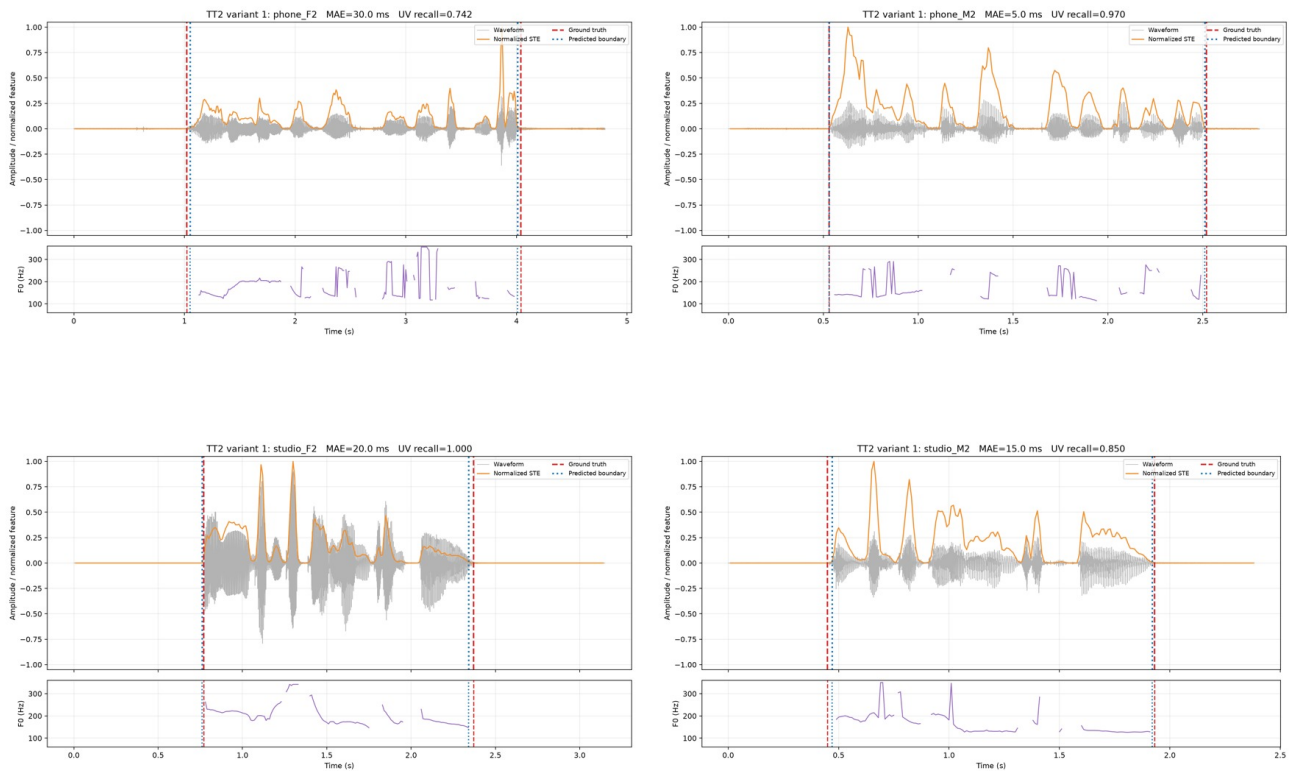


Figure 3. TT2 one-dimensional histogram baseline.

TT2 - DSP extension: four test recordings

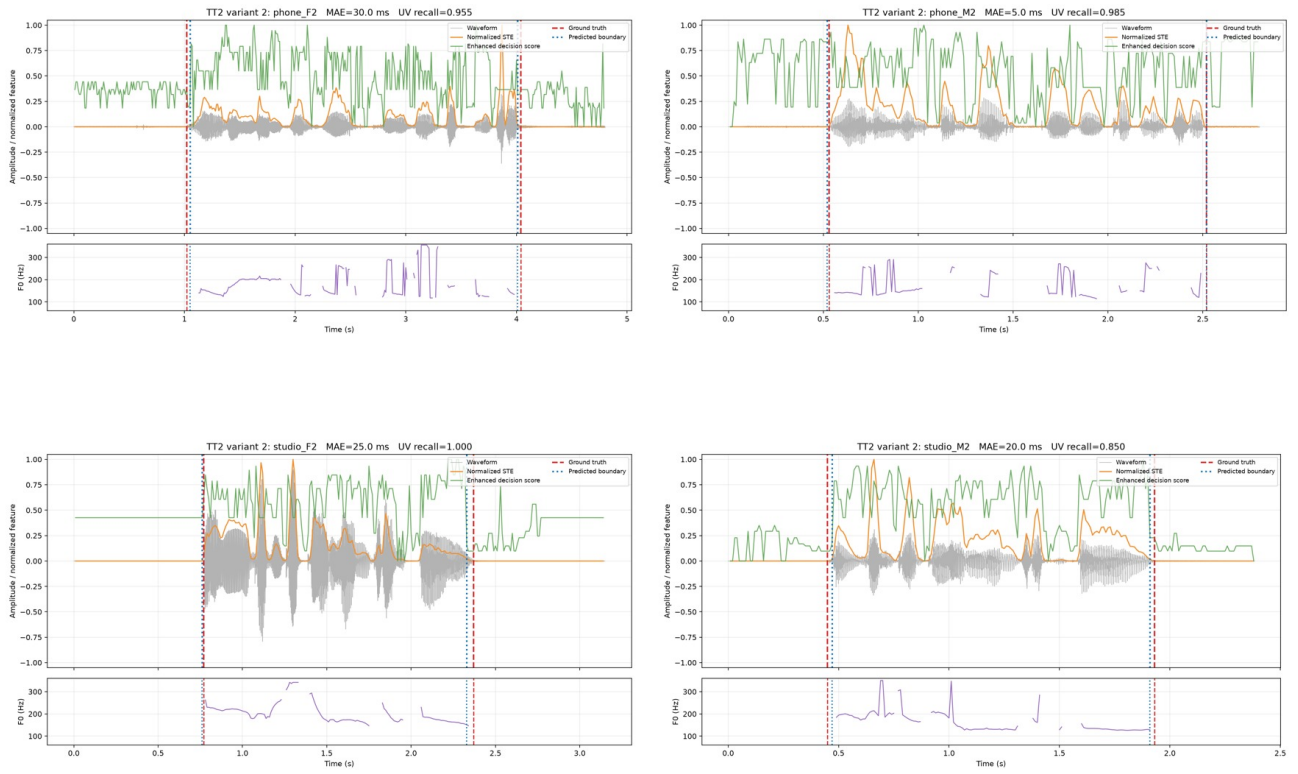


Figure 4. TT2 joint STE-ZCR histogram extension.

The joint histogram increases frame F1 from 0.977 to 0.988 and UV recall from 0.891 to 0.947. Boundary MAE increases from 17.50 ms to 20.00 ms. It preserves more weak unvoiced frames but shifts some outer boundaries by one to three 10 ms hops.

6. TT3 - Gaussian statistical model

6.1 Teacher-aligned baseline

TT3 estimates the mean and standard deviation of normalized STE for speech and silence. Its threshold is the intersection of the two fitted Gaussian densities, using all labeled training frames as required.

6.2 DSP extension

The extension estimates a four-dimensional Gaussian per class over log-STE, pre-emphasized energy, ZCR, and high-frequency ratio. Full covariance matrices model feature correlation; diagonal regularization maintains numerical stability.

TT3 - Baseline: four test recordings

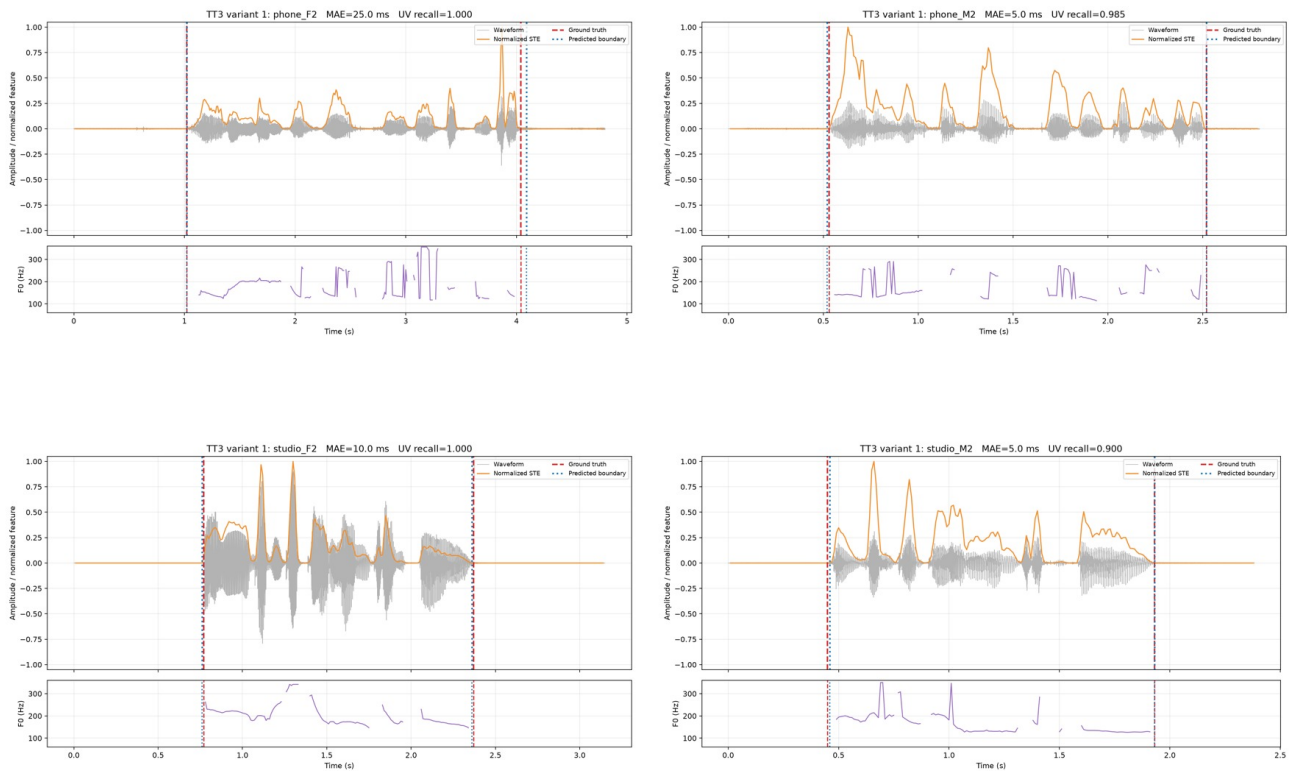


Figure 5. TT3 one-dimensional Gaussian STE baseline.

TT3 - DSP extension: four test recordings

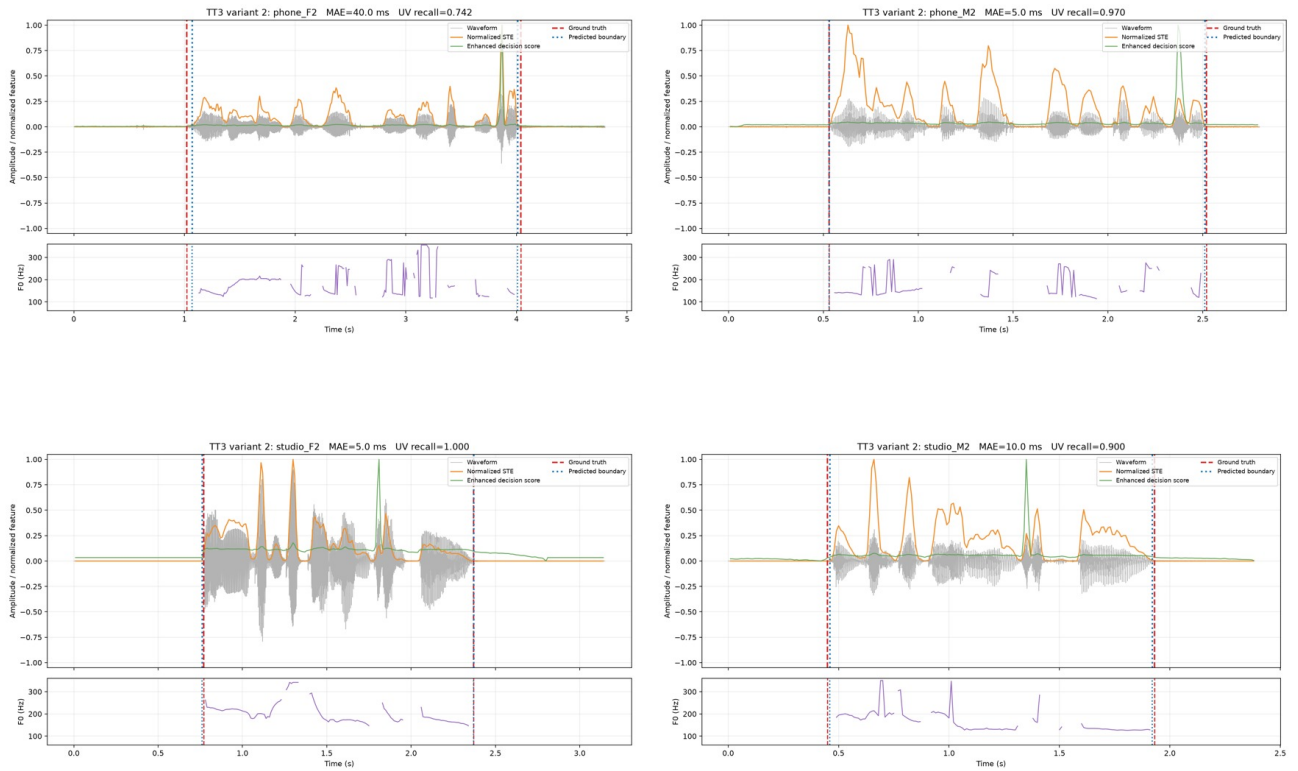


Figure 6. TT3 multivariate Gaussian extension.

The one-dimensional Gaussian obtains 11.25 ms MAE; the multivariate model obtains 15.00 ms. Four training recordings are insufficient for a reliable full covariance estimate, while normalized energy already separates these recordings well. More parameters do not automatically provide better generalization.

7. Quantitative comparison

Method	MAE ms	RMSE ms	Frame F1	UV recall
TT1-1 Global STE threshold	8.75	11.34	0.993	0.971
TT1-2 DSP feature classifier	6.25	7.80	0.996	0.968
TT2-1 1-D STE histogram	17.50	18.81	0.977	0.891
TT2-2 Joint STE-ZCR histogram	20.00	21.56	0.988	0.947
TT3-1 1-D Gaussian STE	11.25	14.87	0.994	0.971
TT3-2 Multivariate Gaussian	15.00	16.34	0.978	0.903

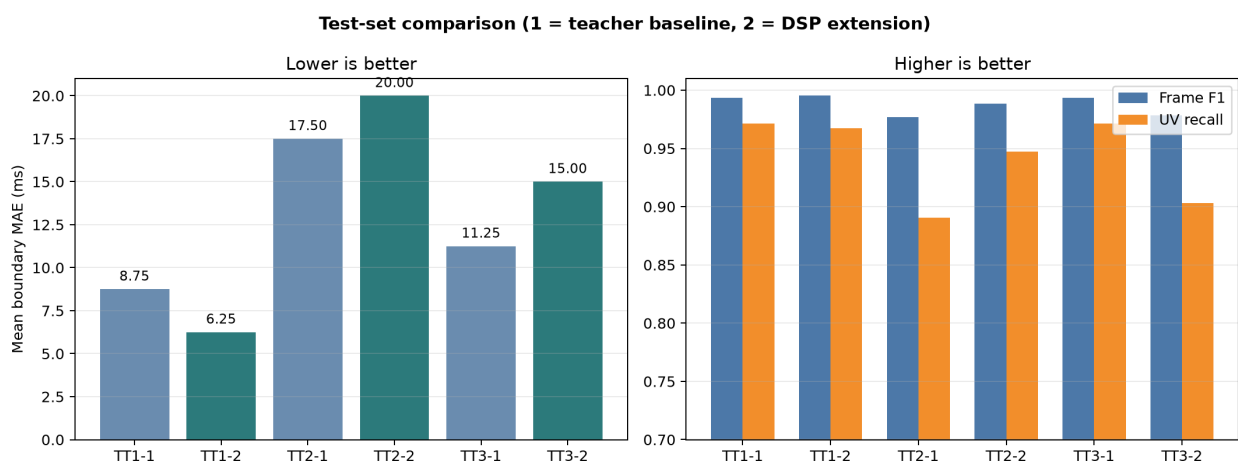


Figure 7. Aggregate test metrics. Variant 1 is the teacher baseline; variant 2 is the DSP extension.

MAE is the assignment's primary boundary metric. RMSE penalizes a large boundary error more strongly. Frame F1 measures the whole speech region. UV recall isolates the known low-energy unvoiced failure mode. Reporting all four prevents one metric from hiding another behavior.

8. Main observations

- **Energy is a strong baseline.** Leading and trailing silence are clear in these recordings.
- **Unvoiced speech is difficult.** It is weaker and more high-frequency/noise-like than voiced speech.
- **The 200 ms rule matters.** It removes short false gaps inside an utterance.
- **More features do not guarantee lower MAE.** TT2 improves UV recall; TT3 overfits covariance.
- **No test tuning was used.** The test set was reserved for the final reported evaluation.

9. Limitations and conclusion

The dataset contains only two environments, two genders, and four training recordings. A larger dataset would support speaker-independent cross-validation and more reliable multivariate density estimates. With the supplied data, TT1-2 is the most accurate tested method at 6.25 ms mean boundary MAE. The main lesson is to add voiced/unvoiced features only when they address a known energy-only failure, and to prefer the simplest model that generalizes.